

Calculus and Linear Algebra

Cramer's Rule and Solution Sets

Lecture 8

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Lecture outline

- ▶ Cramer's rule for square systems with non-zero determinant.
- ▶ Pivot variables, free variables and complete solution sets.
- ▶ Guided 2×3 , 3×2 and 3×3 examples.
- ▶ Compatibility, uniqueness and one or two free parameters.
- ▶ Introduction to systems depending on a coefficient parameter.

Aim

Classify a system and describe **all** its solutions, rather than finding just one example.

Recall: rank determines the number of solutions

For $A \in \mathbb{R}^{m \times n}$, compare the two ranks.

$$\text{rank}(A) < \text{rank}([A \mid b]) \Rightarrow \text{no solution.}$$

If both ranks equal r , the system is compatible:

$$r = n \Rightarrow \text{exactly one solution,}$$

$$r < n \Rightarrow \text{infinitely many solutions.}$$

Today's new step

For a compatible system, the number of free parameters is $n - r$.

Cramer's rule: when can we use it?

Theorem (Cramer) Let $A \in \mathbb{R}^{n \times n}$ with $\det A \neq 0$. The system $Ax = b$ has the unique solution

$$x_i = \frac{\det A_i}{\det A}, \quad i = 1, \dots, n,$$

where A_i is obtained from A by replacing its i th **column** with b .

Conditions

The coefficient matrix must be square and its determinant must be non-zero.

If $\det A = 0$, use ranks and elimination to classify the system.

Which column is replaced?

For three unknowns, write $A = (C_1 \ C_2 \ C_3)$.

$$A_1 = (\textcolor{red}{b} \ C_2 \ C_3), \quad A_2 = (C_1 \ \textcolor{red}{b} \ C_3), \quad A_3 = (C_1 \ C_2 \ \textcolor{red}{b}).$$

$$x = \frac{\det A_1}{\det A}, \quad y = \frac{\det A_2}{\det A}, \quad z = \frac{\det A_3}{\det A}.$$

Keep the order fixed

Column 1 corresponds to x , column 2 to y , and column 3 to z . Replace one column at a time.

Why Cramer's rule works

If $Ax = b$, then $b = x_1 C_1 + \cdots + x_n C_n$. By linearity of the determinant in column i ,

$$\det A_i = \sum_{j=1}^n x_j \det(C_1, \dots, C_{i-1}, C_j, C_{i+1}, \dots, C_n).$$

Every term with $j \neq i$ has two identical columns and vanishes. Hence

$$\det A_i = x_i \det A.$$

Since $\det A \neq 0$, division gives the formula. Existence and uniqueness follow from $\text{rank}(A) = n$.

Cramer for a 2×2 system

Exercise Solve

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

Solution

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad \det A = -2 - 1 = -3 \neq 0.$$

$$A_1 = \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}, \quad \det A_1 = -6, \quad A_2 = \begin{pmatrix} 2 & 5 \\ 1 & 1 \end{pmatrix}, \quad \det A_2 = -3.$$

$$\boxed{x = 2, \quad y = 1.} \quad 2(2) + 1 = 5, \quad 2 - 1 = 1.$$

Cramer for a 3×3 system: setup

Exercise Solve

$$\begin{cases} x + y + z = 6, \\ 2x - y + z = 3, \\ x + 2y + 3z = 14. \end{cases}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 6 \\ 3 \\ 14 \end{pmatrix}.$$

Expanding along the first row,

$$D = \det A = 1(-3 - 2) - 1(6 - 1) + 1(4 + 1) = -5.$$

Cramer's rule applies because $D \neq 0$.

Cramer for a 3×3 system: the numerators

Solution Replace the columns one at a time:

$$A_1 = \begin{pmatrix} 6 & 1 & 1 \\ 3 & -1 & 1 \\ 14 & 2 & 3 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 6 & 1 \\ 2 & 3 & 1 \\ 1 & 14 & 3 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 1 & 6 \\ 2 & -1 & 3 \\ 1 & 2 & 14 \end{pmatrix}.$$

First-row expansions give

$$D_1 = 6(-5) - (9 - 14) + (6 + 14) = -5,$$

$$D_2 = (9 - 14) - 6(6 - 1) + (28 - 3) = -10,$$

$$D_3 = (-14 - 6) - (28 - 3) + 6(4 + 1) = -15.$$

Cramer for a 3×3 system: result and check

$$x = \frac{-5}{-5} = 1, \quad y = \frac{-10}{-5} = 2, \quad z = \frac{-15}{-5} = 3.$$

$$(x, y, z) = (1, 2, 3).$$

Check in the original equations:

$$1 + 2 + 3 = 6, \quad 2 - 2 + 3 = 3, \quad 1 + 4 + 9 = 14.$$

Comparison with Lecture 7

Gaussian elimination gave the same solution. Cramer provides a second method for this non-singular square system.

Choosing a method

Method	Use
Gaussian elimination	Square or rectangular systems, including singular systems. Gives compatibility and complete solutions.
Cramer's rule	Square systems with non-zero determinant. Gives the unique solution.
Minors and ranks	Classification and analysis of exceptional parameter values.

For a

rectangular system, a square subsystem may determine a candidate. The remaining equations must still be checked.

Pivot variables and free variables

Suppose elimination gives

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 5 \\ 0 & 1 & -1 & 1 \end{array} \right).$$

Columns 1 and 2 contain coefficient pivots: x and y are **pivot variables**. Column 3 has no pivot: z is a **free variable**.

$$x + 2z = 5, \quad y - z = 1.$$

Choose z freely, then determine x and y .

Compatibility first

A contradiction row would mean there is no solution, regardless of missing pivots.

One free variable: the complete solution set

Set $z = t$, where $t \in \mathbb{R}$. Then

$$x = 5 - 2t, \quad y = 1 + t, \quad z = t.$$

$$S = \{(5 - 2t, 1 + t, t) : t \in \mathbb{R}\}.$$

Equivalently,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

This is a line in $\mathbb{R}^3$: one point plus one direction.

Why this describes all solutions

For any $t \in \mathbb{R}$, substitution gives

$$(5 - 2t) + 2t = 5, \quad (1 + t) - t = 1.$$

Thus every listed triple is a solution. Conversely, any solution has some real value $z = t$. The equations then force $x = 5 - 2t$ and $y = 1 + t$.

Two checks

A complete description must contain only solutions and must include every solution.

Finding one triple alone does not describe an infinite solution set.

Two free variables

If the only independent equation is

$$x + y + z = 3,$$

choose $y = s$ and $z = t$, independently. Then $x = 3 - s - t$.

$$S = \{(3 - s - t, s, t) : (s, t) \in \mathbb{R}^2\}.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

The two direction vectors are independent, so the solution set is a plane.

Counting parameters and understanding the geometry

For a compatible system with n unknowns and rank r ,

number of free parameters = $n - r$.

With three unknowns:

Rank	Free parameters	Solution set
3	0	One point
2	1	A line
1	2	A plane
0	3	All of $\mathbb{R}^3$

These conclusions require compatibility.

All possible cases for the requested dimensions

Size	$r = \text{rank}(A)$	Compatible: free parameters	Incompatible when
2×3	2	1	Impossible: augmented rank cannot exceed 2
	1	2	Augmented rank is 2
	0	3	$b \neq 0$
3×2	2	0: unique	Augmented rank is 3
	1	1	Augmented rank is 2
	0	2	$b \neq 0$
3×3	3	0: unique	Impossible
	2	1	Augmented rank is 3
	1	2	Augmented rank is 2
	0	3	$b \neq 0$

A 2×3 system with one free parameter

Exercise Classify and solve

$$\begin{cases} x + y + z = 2, \\ 2x - y + z = 1. \end{cases}$$

Use $R_2 \mapsto R_2 - 2R_1$:

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -3 & -1 & -3 \end{array} \right).$$

Both ranks equal $2 < 3$. The system has one free variable. The pivots are in the x and y columns, so choose $z = t$.

The 2×3 system: complete solution and check

Solution From $-3y - z = -3$ and $x + y + z = 2$,

$$z = t, \quad y = 1 - \frac{t}{3}, \quad x = 1 - \frac{2t}{3}.$$

$$S = \left\{ \left(1 - \frac{2t}{3}, 1 - \frac{t}{3}, t \right) : t \in \mathbb{R} \right\}.$$

Check for every t :

$$1 - \frac{2t}{3} + 1 - \frac{t}{3} + t = 2,$$

$$2 \left(1 - \frac{2t}{3} \right) - \left(1 - \frac{t}{3} \right) + t = 1.$$

A 2×3 system with two free parameters

Exercise Solve

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 6. \end{cases}$$

With $R_2 \mapsto R_2 - 2R_1$,

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Both ranks are 1, so $n - r = 3 - 1 = 2$. **Solution** Set $y = s$, $z = t$. Then

$$(x, y, z) = (3 - s - t, s, t), \quad (s, t) \in \mathbb{R}^2.$$

An incompatible 2×3 system

Exercise Classify

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 7. \end{cases}$$

Solution Elimination gives

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{array} \right).$$

The ranks are 1 and 2, and the second row is $0 = 1$.

$$\boxed{S = \emptyset.}$$

Having more unknowns than equations does not guarantee compatibility.

A 3×2 system with a unique solution

Exercise Solve

$$\begin{cases} x + y = 3, \\ 2x - y = 0, \\ 3x = 3. \end{cases}$$

Use $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - 3R_1$, then $R_3 \mapsto R_3 - R_2$:

$$[A \mid b] \sim \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{array} \right).$$

Both ranks are $2 = n$. Back substitution gives

$$\boxed{x = 1, \quad y = 2.}$$

A square subsystem: Cramer and the remaining equation

The first two equations form a square subsystem:

$$D = \det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = -3, \quad D_x = \det \begin{pmatrix} 3 & 1 \\ 0 & -1 \end{pmatrix} = -3, \quad D_y = \det \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix} = -6.$$

Cramer gives the candidate $x = 1, y = 2$. The third equation holds because $3(1) = 3$.

Every equation matters

If the third equation were $3x = 4$, the same candidate would fail and the full system would be incompatible.

An incompatible 3×2 system of full column rank

Consider

$$\begin{cases} x + y = 3, \\ 2x - y = 0, \\ 3x = 4. \end{cases}$$

The same row operations now give

$$[A \mid b] \sim \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 1 \end{array} \right).$$

$$\text{rank}(A) = 2, \quad \text{rank}([A \mid b]) = 3.$$

Thus $S = \emptyset$, even though both coefficient columns are independent.

A 3×2 system with one free parameter

Exercise Solve

$$\begin{cases} x + 2y = 4, \\ 2x + 4y = 8, \\ -x - 2y = -4. \end{cases}$$

Use $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 + R_1$:

$$[A \mid b] \sim \left(\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right).$$

Both ranks equal 1. Set $y = t$, giving

$$S = \{(4 - 2t, t) : t \in \mathbb{R}\}.$$

A 3×2 system: dependent coefficients can contradict

Change the third constant:

$$\begin{cases} x + 2y = 4, \\ 2x + 4y = 8, \\ -x - 2y = -3. \end{cases}$$

Then $R_3 \mapsto R_3 + R_1$ produces $0 = 1$.

$$[A \mid b] \sim \left(\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right).$$

Coefficient rank 1, augmented rank 2: no solution. Proportional coefficients require the same proportionality in the constants.

Two free parameters in a 3×2 system

For two unknowns, two free parameters require $r = 0$. Thus every coefficient must be zero.

$$\begin{cases} 0x + 0y = 0, \\ 0x + 0y = 0, \\ 0x + 0y = 0. \end{cases}$$

Every pair satisfies the equations:

$$S = \{(s, t) : (s, t) \in \mathbb{R}^2\} = \mathbb{R}^2.$$

If even one right-hand side is non-zero, the system instead has no solution. This is the only compatible 3×2 case with two free parameters.

A 3×3 system with a unique solution: exercise

Exercise Solve by elimination:

$$\begin{cases} x + y + z = 4, \\ 2x + 3y + z = 9, \\ x + 2y + 3z = 9. \end{cases}$$

1. Find an echelon augmented matrix.
2. Explain compatibility and uniqueness using both ranks.
3. Find the solution and verify all three equations.

The unique 3×3 solution: elimination

Solution Apply $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$:

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 1 & 2 & 5 \end{array} \right).$$

Then $R_3 \mapsto R_3 - R_2$ gives

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 3 & 4 \end{array} \right).$$

Both ranks are $3 = n$, so the solution is unique.

The unique 3×3 solution: back substitution

$$z = \frac{4}{3}, \quad y = 1 + z = \frac{7}{3}, \quad x = 4 - y - z = \frac{1}{3}.$$

$$(x, y, z) = \left(\frac{1}{3}, \frac{7}{3}, \frac{4}{3} \right).$$

Check in the original system:

$$\frac{1 + 7 + 4}{3} = 4, \quad \frac{2 + 21 + 4}{3} = 9, \quad \frac{1 + 14 + 12}{3} = 9.$$

The fraction-valued solution is valid: solutions need not be integers.

A 3×3 system with one free parameter

Exercise Solve

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 6, \\ x - y + z = 1. \end{cases}$$

Use $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - R_1$, then swap rows 2 and 3:

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Both ranks are 2. There is one free variable, z .

One free parameter: complete solution and geometry

Solution The second row gives $y = 1$. The first gives $x + z = 2$. Set $z = t$:

$$S = \{(2 - t, 1, t) : t \in \mathbb{R}\}.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Check: $(2 - t) + 1 + t = 3$, twice this is 6, and $(2 - t) - 1 + t = 1$. The three planes have a common intersection line.

A 3×3 system with two free parameters

Exercise Solve

$$\begin{cases} x + 2y - z = 4, \\ 2x + 4y - 2z = 8, \\ -x - 2y + z = -4. \end{cases}$$

Eliminate with $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 + R_1$:

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Both ranks are 1, so $n - r = 2$.

Two free parameters: complete solution and check

Solution Choose $y = s$ and $z = t$ independently:

$$S = \{(4 - 2s + t, s, t) : (s, t) \in \mathbb{R}^2\}.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Substitution gives $(4 - 2s + t) + 2s - t = 4$. The other two equations are twice and minus this equation. The solution set is a plane in $\mathbb{R}^3$.

An incompatible 3×3 system with rank 2

Consider

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 7, \\ x - y + z = 1. \end{cases}$$

The same operations used in the one-parameter example give

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right).$$

The ranks are 2 and 3. The last row is $0 = 1$.

$$\boxed{S = \emptyset.}$$

An incompatible 3×3 system with rank 1

Consider

$$\begin{cases} x + 2y - z = 4, \\ 2x + 4y - 2z = 9, \\ -x - 2y + z = -4. \end{cases}$$

Elimination and a row ordering give

$$[A \mid b] \sim \left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Coefficient rank 1, augmented rank 2: no solution.

Rank deficiency alone

A low coefficient rank yields free parameters only when the constants are compatible.

The zero coefficient matrix: all dimensions

If $A = 0$, then $Ax = b$ becomes $0 = b$.

- ▶ If $b = 0$, every $x \in \mathbb{R}^n$ is a solution. There are n free parameters.
- ▶ If $b \neq 0$, there is no solution. The two ranks are 0 and 1.

This completes the edge cases

For 2×3 or 3×3 , the compatible rank-zero case has three free parameters. For 3×2 , it has two.

Homogeneous systems and linear dependence

For $Ax = 0$, the system is always compatible.

$$\text{rank}([A \mid 0]) = \text{rank}(A) = r.$$

- ▶ $r = n$: the only solution is the zero vector.
- ▶ $r < n$: there are $n - r$ free parameters and non-zero solutions.

Connection with columns

A non-zero solution to $Ax = 0$ gives a non-trivial linear dependence relation among the columns of A .

A homogeneous 2×3 example

Exercise Solve

$$\begin{cases} x + y + z = 0, \\ 2x - y + z = 0. \end{cases}$$

Elimination gives $-3y - z = 0$. Choose $z = 3t$ to avoid fractions:

$$y = -t, \quad x = -y - z = -2t.$$

$$S = \{t(-2, -1, 3) : t \in \mathbb{R}\}.$$

Check: $-2 - 1 + 3 = 0$ and $2(-2) - (-1) + 3 = 0$. The solution line passes through the origin.

Different parameterizations can describe the same set

For the previous example, both expressions describe all solutions:

$$\left(-\frac{2u}{3}, -\frac{u}{3}, u\right), \quad u \in \mathbb{R},$$

$$(-2t, -t, 3t), \quad t \in \mathbb{R}.$$

They agree under the change $u = 3t$.

What matters

The chosen free variable or parameter name may differ. The set of solution vectors must be the same.

Two meanings of the word parameter

Free solution parameter

$$(x, y) = (4 - 2t, t).$$

t labels different solutions of **one fixed system**.

Coefficient parameter

$$\begin{cases} x + y = 2, \\ x + \lambda y = 3. \end{cases}$$

λ changes the system itself. For each fixed λ , we solve a different system.

A special coefficient value may create free solution parameters or make the system incompatible.

Systems depending on a coefficient parameter

Consider a family

$$A(\lambda)x = b(\lambda), \quad \lambda \in \mathbb{R}.$$

Our task is to classify and solve the system for **every** value of λ .

1. Identify values where a determinant or pivot vanishes.
2. Solve the generic case where the required quantities are non-zero.
3. Substitute each exceptional value into the original system.
4. Compare ranks and solve each exceptional case separately.

A parameter-dependent 2×2 system

Consider

$$\begin{cases} x + y = 2, \\ x + \lambda y = 3. \end{cases}$$

$$\det A(\lambda) = \det \begin{pmatrix} 1 & 1 \\ 1 & \lambda \end{pmatrix} = \lambda - 1.$$

Elimination gives

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & \lambda & 3 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & \lambda - 1 & 1 \end{array} \right).$$

The value $\lambda = 1$ must be examined separately.

Generic and exceptional values

If $\lambda \neq 1$, there is a unique solution:

$$y = \frac{1}{\lambda - 1}, \quad x = 2 - \frac{1}{\lambda - 1} = \frac{2\lambda - 3}{\lambda - 1}.$$

If $\lambda = 1$, the original equations become

$$x + y = 2, \quad x + y = 3.$$

They contradict each other, so there is no solution.

The restriction is part of the answer

The rational formulas apply only for $\lambda \neq 1$. A vanishing denominator requires a separate analysis.

A different right-hand side changes the exceptional case

Now consider

$$\begin{cases} x + y = 2, \\ x + \lambda y = 2. \end{cases}$$

Elimination yields $(\lambda - 1)y = 0$.

- ▶ If $\lambda \neq 1$, then $y = 0$, $x = 2$: one solution.
- ▶ If $\lambda = 1$, the equations coincide. Both ranks are 1 and

$$(x, y) = (2 - t, t), \quad t \in \mathbb{R}.$$

The same coefficient determinant can lead to different exceptional outcomes depending on b .

Preview for Lecture 9: all three outcomes in one family

Consider

$$\begin{cases} x + y + z = 3, \\ x + (1 + \lambda)y + z = 3 + \lambda, \\ x + y + (\lambda^2 - 1)z = 4. \end{cases}$$

Subtracting the first equation from the other two gives

$$\lambda y = \lambda, \quad (\lambda^2 - 2)z = 1.$$

Exercise Which values of λ need separate treatment? In Lecture 9 we will develop the complete case analysis and solutions.

Preview: classification of the family

The coefficient determinant is $\lambda(\lambda^2 - 2)$.

- ▶ $\lambda \notin \{0, \sqrt{2}, -\sqrt{2}\}$: three coefficient pivots, hence a unique solution.
- ▶ $\lambda = 0$: $0 = 0$ and $-2z = 1$. Both ranks are 2, so there is one free parameter.
- ▶ $\lambda = \pm\sqrt{2}$: the last equation reduces to $0 = 1$, so there is no solution.

Lecture 9

We will compute the complete solutions and check each case in the original equations.

Common mistakes with parameters

- ▶ Dividing by an expression in λ without recording when it is non-zero.
- ▶ Using Cramer's formulas when the determinant vanishes.
- ▶ Treating a coefficient parameter as a freely chosen solution variable.
- ▶ Giving an infinite solution set without stating the parameter ranges.
- ▶ Counting parameters before checking compatibility.
- ▶ Checking only a square subsystem of an overdetermined system.

Closing the linear systems unit

- ▶ Matrix and augmented forms, row operations and echelon form.
- ▶ Rank, pivots and the Rouché–Capelli theorem.
- ▶ Unique, infinite and empty solution sets, including rectangular systems.
- ▶ Free variables, parametric descriptions and homogeneous systems.
- ▶ Cramer's rule and its conditions.
- ▶ Lecture 9: coefficient parameters, exceptional values and mixed practice.

After Lecture 9

This completes the core methods for linear systems. We can then move to functions, domains and graphs as the starting point for Calculus.

Independent practice: one or two free parameters?

Exercise Classify and describe the complete solution sets:

$$\text{A: } \begin{cases} x + y + z = 1, \\ x - y + z = 3, \end{cases} \quad \text{B: } \begin{cases} x - y + 2z = 2, \\ 2x - 2y + 4z = 4. \end{cases}$$

For each system, state both ranks, choose free variables, and verify your result for arbitrary parameter values. **C:** What changes in B if the second constant becomes 5?

Independent practice: full solutions

Solution A: subtract the first equation from the second to get $-2y = 2$, so $y = -1$. Then $x + z = 2$.

$$S_A = \{(2 - t, -1, t) : t \in \mathbb{R}\}, \quad r_A = r_{[A|b]} = 2.$$

B: the second equation is twice the first. Set $y = s$, $z = t$:

$$S_B = \{(2 + s - 2t, s, t) : (s, t) \in \mathbb{R}^2\}, \quad r = 1.$$

Checks: $(2 - t) - 1 + t = 1$ and $(2 - t) + 1 + t = 3$. For B, $(2 + s - 2t) - s + 2t = 2$.

C: subtracting twice the first equation gives $0 = 1$. The ranks are 1 and 2, so no solution.

Independent practice: a 3×2 system

Exercise Classify and solve

$$\begin{cases} x - y = 1, \\ 2x + y = 5, \\ 3x = 6. \end{cases}$$

Then replace the third equation by $3x = 7$ and classify the new system. Use either elimination or Cramer on the first two equations, followed by a check of the third equation.

Independent practice: 3×2 solution

Solution Apply $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - 3R_1$, then $R_3 \mapsto R_3 - R_2$:

$$[A \mid b] \sim \left(\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & 3 \\ 0 & 0 & 0 \end{array} \right).$$

Both ranks equal 2. Thus $y = 1$, $x = 2$. Checks: $2 - 1 = 1$, $4 + 1 = 5$, $3(2) = 6$. If the third constant is 7, the last row becomes $(0 \ 0 \mid 1)$. The ranks are 2 and 3, so the new system has no solution.

Independent practice: Cramer

Exercise Solve by Cramer's rule:

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

Solution

$$D = \det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = -3,$$

$$D_x = \det \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix} = -6, \quad D_y = \det \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix} = -9.$$

Therefore $x = 2$, $y = 3$. Check: $2 + 3 = 5$ and $4 - 3 = 1$.